



Empowering Learners to Build Production - Ready
AI Agents

Applied Agentic AI for EMs

Week 0 - Exercise

Problem Statement

**BankCo — Retaining Premium Credit Card
Customers with an Internal Agent**

What is this document for?

Purpose: Introduce a real-world use case to help you apply key **Agentic AI concepts** before starting the course.

Focus areas:

- Decision support
- Process automation
- Personalized recommendations

Scenario: Explore how AI agents can **analyze data, make decisions, and recommend actions** to improve **business processes and customer retention** for a bank's premium credit card customers.

What you'll do:

- Identify **signals of at-risk customers**
- Design a **basic AI agent workflow**
- Consider **how to communicate and implement solutions**

Approach:

- A **thought exercise** — no coding required
- Focus on **how generative AI systems work in practice** and **where human decision-making fits**

Goal: Spark **curiosity and practical thinking** about how AI can be used in your role.

BankCo — Retaining Premium Credit Card Customers with an Internal Agent

1. Product and Business Context

Company: BankCo (a large bank)

Product in Focus: Premium credit card

- Annual fee
- Airport lounge access
- Concierge services
- Higher rewards

Primary Stakeholders:

- Relationship Managers (RMs)
- Retention Operations team (small)

2. The Challenge

Problem Statement

“Identify premium customers who are 30–90 days away from their annual fee renewal and show signs of disengagement or dissatisfaction, then prepare Relationship Manager-ready recommendations to prevent cancellations or downgrades.”

What Goes Wrong Today

- RMs **lack a single view** of who is at risk and why.
- RMs **are overloaded and rely on manual processes.**
 - Manual lists
 - Generic email templates
 - Outreach is often late or mismatched to customer needs

Business Impact (Why We Care)

- **Preventable cancellations** near renewal reduce customer lifetime value and hurt NPS.
- **Poor timing and generic messaging** waste RM time and retention budget.
- **Inconsistent offer application** introduces compliance and fairness risk.
- **Premium customers are expensive to acquire.**

If they leave at renewal, BankCo loses:

 - Annual fee revenue
 - Spending volume
 - Future cross-sell potential
- **Competitors are actively poaching premium customers** with better perks or timely offers.

Goal of This Exercise

Design **one internal AI agent** that helps RMs **retain at-risk premium customers** who are approaching renewal.

3. What “Agent” Means Here (Week 0 Scope)

For this exercise, an **agent** is an internal AI assistant that can:

- Read multiple connected spreadsheets (as separate tabs)
- Identify at-risk customers using the provided signals (or your improved version)
- Select a compliant retention offer using the Offer Policy tab
- Generate RM-ready artifacts:
 - Prioritized customer list
 - Personalized email draft per customer
- Write recommendations into a tracking sheet used in the RM workflow
- Notify the RM by email that recommendations are ready

Important Boundaries

- The agent does not contact customers directly; RMs remain fully in control of outreach.
- The agent does not change customer settings (no fee waivers, tier changes, or credit limit changes).
- The agent provides decision support and drafting only, not execution.

4. Data and System Assumptions

Dataset - Week0 Data Set

Assume all data is **updated daily**.

All communication is via **email**.

5) “Risk signals” you can use (and you can change them)

To keep this warm-up practical, assume BankCo starts with the following **baseline rules** to shortlist at-risk customers. You may use these as-is, adjust thresholds, add signals, or propose a different approach.

Step 1 — Eligibility filter (required)

Customer is **in scope** if:

- **annual_fee_due_date** is **30–90 days** from today.

Step 2 — At-risk rules

Flag a customer as **at-risk** if they meet **either** condition:

A) “2-of-N” rule (general risk): Any 2 of the signals below are true.

Engagement decline / benefit underuse

- `total_spend_90d` is low (threshold: `< 4000`)
- `travel_share_change` `<= -0.20`
- `lounge_visits_90d` `= 0`

Low value realization

- `redemptions_90d` `= 0`

Negative experience

- `complaints_90d` `>= 1`
- `disputes_90d` `>= 1`
- `benefit_denials_90d` `>= 1`
- `nps_last` `<= 0`

OR

B) “Severe signal” rule (immediate risk): Any **one** of these triggers is enough:

- `complaints_90d` `>= 2`
- `nps_last` `<= -20`
- `benefit_denials_90d` `>= 2`

Notes for learners:

- The numbers above are **thresholds**, not “correct answers.”
- You can argue for different thresholds, use relative thresholds (e.g., bottom quartile spend), or add new heuristics (e.g., “high balance + low spend” as fee sensitivity).
- Your goal is to show sound product reasoning, not perfect modeling.

6. Your Tasks

You are **not building the agent yet**.

Your task is to **demonstrate design thinking**.

Step 1: Clarify the Goal

Write **2–3 sentences** covering:

- The **RM's user problem**
- The **BankCo business goal**

Then select:

- **Primary Success metric**
- **Secondary Success metric**

Step 2: Identify the Signals That Matter

From the tabs above:

- Select 5 key fields used to detect risk and/or tailor offers and messaging
- For each field, add one sentence explaining why it matters

You may use the provided risk signals as-is or propose changes.

Step 3: Define the Agent Workflow

Write **6–8 sequential steps** describing:

- How customers enter the at-risk set (renewal window + signals)
- How risk reasons are generated
- How eligible offers are selected using Offer Policy
- How email drafts are created
- How recommendations are logged in Retention_Recommendations_Log
- How RMs are notified by email

Step 4: Draw the System Sketch

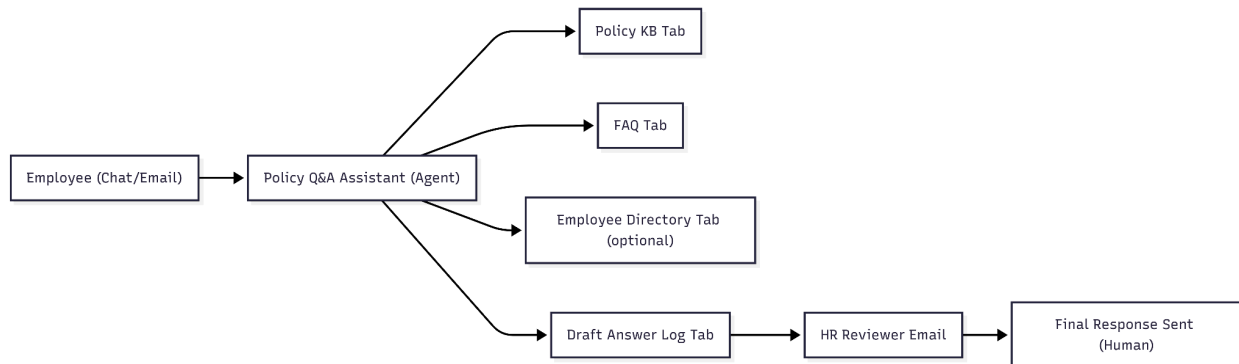
Include these components:

- RMs
- Premium Customers
- Spend Signals
- Experience Signals
- Offer Policy
- Premium Retention Assistant (Agent)
- Retention Recommendations Log
- RM Email Notification

Show **arrows indicating data flow** (text diagram is sufficient).

For Example -

Use case: Employees ask questions about leave/expense policy. The agent drafts an answer grounded in internal policy tables. HR reviews and sends (agent does not auto-send).



Out of Scope for Week 0

You do **not** need to worry about:

- Selecting AI models or writing prompts
- Legal or compliance edge cases
- Building the agent

Measuring downstream customer outcomes